Leeds School of Business
UNIVERSITY OF COLORADO BOULDER

MBAX 6418 – Fall 2026
Building Business Solutions with Generative AI & LLMs

Course Meeting Information

Days	Time	Venue
Mondays	09:30 AM to 12:15 PM	WLAW 304
Thursdays	08:00 AM to 10:45 AM	KOBL 217

Professor Information

Professor: David Dobolyi, PhD
Email: david.dobolyi@colorado.edu
Office: KOBL 401F

Office Hours: Tuesdays and Fridays 11am to 12pm via Zoom (https://cuboulder.zoom.us/my/dobolyi), and by appointment
Please notify me of your intention to come by, either in class or via email. ***I am happy to schedule appointments at other times*** *based on availability.*

Teaching Assistant (TA) Information

TAs: None

Course Description

In this course, students will learn how to leverage recent advancements in generative artificial intelligence (GAI) to solve business problems. The course focuses on three key aspects of GAI for business. First, students will learn how best to use open and closed large language models (LLMs) for coding, including topics such as AI-assisted coding, vibe coding, and debugging. Next, students will develop their skills with models that focus on text as well as multimodal models incorporating text, vision, and/or audio (e.g., text to image models): Relevant topics include fine-tuning, sustainable model deployment, and agentic workflows. Finally, students will learn how to quickly and easily share their models with stakeholders through a variety of methods and platforms, including graphical user interfaces (GUIs), application programming interfaces (APIs), and local/cloud hosting options. Coding experience is not a prerequisite for this course.

Course Objectives

- Survey the current state-of-the-art (SOTA) in open and closed GAI and LLMs
- Explore use cases for various model types, e.g., code-focused, multimodal, and reasoning models
- Become familiar with key GAI concepts such as model context protocol (MCP)
- Learn how to use AI coding agents effectively
- Fine-tune models to develop bespoke solutions to business challenges

- Evaluate model prediction outcomes to measure success and understanding
- Optimize model deployments for efficient inferencing and sustainable serving
- Work through business problems from start to finish using GAI and LLM technology
- Communicate findings to non-technical audiences

Textbooks & Course Materials

- Handouts (e.g., slides) and other course material distributed in class or via Canvas and GitHub
- Open-Source Software (e.g., Hermes Agent, Visual Studio Code)
- Open-Source Python Packages (e.g., Transformers, vLLM, SGLang, Unsloth, LangChain, LlamaIndex, OpenAI Python API, Streamlit, Gradio, JupyterLab)
- Cloud Platforms (e.g., Google Colab, OpenAI Developer Platform, Google AI Studio)
- Commercial AI Subscriptions (e.g., ChatGPT, Claude)

Note: The bulk of the course materials are free, but students may need to acquire a paid subscription for services such as ChatGPT (https://chatgpt.com/pricing/) depending on their usage and specific needs. That said, I will be providing free API endpoints running local AI models for students for this class.

Canvas

I will use the Canvas system, which can be accessed at: https://canvas.colorado.edu/. I will post the syllabus, course materials, slides, etc. there. In addition, I will use Canvas for course announcements as needed (e.g., schedule updates/changes, class cancellations in the case of inclement weather, etc.).

Course Grading

Final grades will be determined as follows:

Item	Percentage
Assignments	20%
Quizzes	15%
Final Project	45%
Final Exam	15%
Attendance and Participation	5%
	100%

The instructor reserves the right to grade on the curve.

Assignments (20%)

Students will complete assignments throughout the semester related to topics covered in class, and these assignments may be either individual or team-based. It is critical that you complete (or at least attempt) every assignment, since they make up a large portion of your class grade. Not submitting assignments on time will result in a 10% grade penalty for every calendar day the assignment is late (e.g., an assignment that would have received a grade of 80/100 if turned in on time would receive a grade of 72/100 [.80 * .90 = .72] if turned in up to one day late or 64/100 [.80 * .80 = .64] if turned up to

two days late [and so on]). Assignments may not be turned in more than five days late (e.g., so that we may review in class and/or so that answer keys may be provided to other students in a timely fashion), but the assignment with the lowest score will be dropped from your final grade.

Quizzes (15%)

Quizzes will be given during class for in-person students or made available on Canvas for online students, and the dates for these quizzes will be announced in advance. These quizzes are intended to ensure that students understand core concepts covered during prior classes. Quizzes must be completed on time to receive course credit: I will **NOT** accept late quiz submissions, but the quiz with the lowest score will be dropped from your final grade.

Final Project (45%)

The team-based final project is designed to give you practical, hands-on experience in designing a business solution using the tools and techniques you learn about during this course. The project will begin with a proposal and culminate in a brief presentation that provides a broad overview of your work. More information about the project (e.g., requirements, rubrics) will be made available on Canvas about one month into the course.

Final Exam (15%)

There will be a final exam at the end of the course, which will be broadly similar in terms of overall content to class assignments. The exam will be completed remotely (instructions will be provided as the exam nears).

Attendance and Participation (5%)

The objective of class attendance and participation is to provide a means of assessing the degree to which individuals contribute to the learning of their peers. Students who show up on time, are prepared for class, and regularly contribute to the conversation in a positive way will receive full credit. **However, negative behaviors will result in a *significant reduction or complete forfeiture* of this portion of your grade.** Additionally, sign-in sheets may occasionally be used to formally assess in-person attendance.

In-Person Students. Attendance and participation for in-person students will be assessed based on the following dimensions:

1. **Attendance:** Class attendance is extremely important to succeed in the course. You are expected to be in class and prepared to engage actively with the lecture and discussions. Participation is negatively impacted when you arrive late for class or miss class. **Any observed unexcused absence beyond the first will result in a reduction of your class participation grade.** It is generally not possible to "make up" these absences with extra credit or other arrangements, so please plan your schedule carefully and be mindful of the attendance policy to avoid problems.
2. **Technology Distractions: Unless expressly permitted, cell phones are not permitted to be used in class at any time for any reason.** If you have an emergency situation that necessitates your being available for a phone call, let me know before class. Due to the technical nature of this class, we

will occasionally utilize laptops during lectures. However, you are expected to stay on-topic and refrain from checking email, chatting, or web surfing unrelated topics. **Disruptions that affect others may result in a reduction of your class participation grade.**

Online Students. Attendance and participation for online students will primarily be based on completing a series of short "check-in" surveys throughout the semester that will be made available on Canvas. In addition, for online programs that include synchronous sessions (e.g., via Zoom), attendance and participation related to those sessions will also factor into this grading component.

Classroom Behavior

Students and faculty are responsible for maintaining an appropriate learning environment in all instructional settings, whether in person, remote, or online. Failure to adhere to such behavioral standards may be subject to discipline. Professional courtesy and sensitivity are especially important with respect to individuals and topics dealing with race, color, national origin, sex, pregnancy, age, disability, creed, religion, sexual orientation, gender identity, gender expression, veteran status, marital status, political affiliation, or political philosophy.

Additional classroom behavior information

- Student Classroom and Course-Related Behavior Policy.
- Student Code of Conduct.
- Office of Institutional Equity and Compliance.

Accommodation for Disabilities, Temporary Medical Conditions, and Medical Isolation

If you qualify for accommodations because of a disability, please submit your accommodation letter from Disability Services to your faculty member in a timely manner so that your needs can be addressed. Disability Services determines accommodations based on documented disabilities in the academic environment. Information on requesting accommodations is located on the Disability Services website. Contact Disability Services at 303-492-8671 or DSinfo@colorado.edu for further assistance. If you have a temporary medical condition, see Temporary Medical Conditions on the Disability Services website.

If you have a temporary medical condition or required medical isolation for which you require accommodation, please reach out to me via email at david.dobolyi@colorado.edu. Note that you do not need to state the nature of your illness when reaching out to me, and you also do not need to supply a "doctor's note" for missed classes, etc.

Student Names and Pronouns

CU Boulder recognizes that students' legal information does not always align with how they identify. If you wish to have a name other than your legal name appear on your instructors' class rosters and in Canvas, or if you wish to choose pronouns to appear on your instructors' class rosters and in Canvas, visit the Registrar's website for instructions on how to change your personal information in university systems.

Honor Code

All students enrolled in a University of Colorado Boulder course are responsible for knowing and adhering to the Honor Code. Violations of the Honor Code may include but are not limited to: plagiarism (including use of paper writing services or technology [such as essay bots]), cheating, fabrication, lying, bribery, threat, unauthorized access to academic materials, clicker fraud, submitting the same or similar work in more than one course without permission from all course instructors involved, and aiding academic dishonesty. Understanding the course's syllabus is a vital part of adhering to the Honor Code.

All incidents of academic misconduct will be reported to Student Conduct & Conflict Resolution: StudentConduct@colorado.edu. Students found responsible for violating the Honor Code will be assigned resolution outcomes from Student Conduct & Conflict Resolution and will be subject to academic sanctions from the faculty member. Visit Honor Code for more information on the academic integrity policy.

Sexual Misconduct, Discrimination, Harassment and/or Related Retaliation

CU Boulder is committed to fostering a productive and welcoming learning, working, and living environment. University policy prohibits protected-class discrimination and harassment, sexual misconduct (harassment, exploitation, and assault), intimate partner abuse (dating or domestic violence), stalking, and related retaliation by or against members of our community on- or off-campus. Denial of an approved accommodation for disability, religious observance, or pregnancy or pregnancy related medical conditions may be discriminatory.

The Office of Institutional Equity and Compliance (OIEC) addresses these concerns, and individuals who have been subjected to misconduct can contact OIEC at 303-492-2127 or email OIEC@colorado.edu. Information about university policies, OIEC reporting options, and OIEC support resources including confidential services can be found on the OIEC website.

Faculty and graduate instructors are required to inform OIEC when someone discloses misconduct regardless of when or where it occurred. This is to ensure that those impacted receive outreach from OIEC about resolution options and support resources. To learn more about reporting and support options for a variety of concerns, visit the Don't Ignore It page.

Accommodation for Religious Obligations

Instructional faculty members must make every reasonable effort to accommodate all students who have conflicts with scheduled exams, assignment deadlines or required attendance due to a religious observance. Whenever possible, students must notify the instructional faculty member at least two weeks in advance of the expected exam, assignment deadline, or attendance conflict to request an accommodation for religious observance. If the start date of the course is less than two weeks before the date of requested accommodation, students must notify the instructional faculty member on the start date of the course. See the Student Academic Accommodations for Religious Observances Policy for more information.

In this class, please reach out to me via email at david.dobolyi@colorado.edu with any questions or concerns you may have in a timely manner.

Mental Health and Wellness

The University of Colorado Boulder is committed to supporting students' mental health and overall wellbeing. If personal, academic, or emotional challenges are affecting your wellbeing or success, Counseling and Psychiatric Services (CAPS), is here to help. CAPS offers counseling, referrals, psychiatric care, crisis support, and much more. Visit CAPS in the C4C, or call (303) 492-2277, 24/7.

Acceptable Use of AI in This Class

You may use gen AI tools in this course however you see fit. In fact, some assignments will require the use of such tools. However, keep in mind your own and others' privacy implications and security risks when using gen AI tools. Document your usage by following the guidelines provided during the course.

TENTATIVE COURSE SCHEDULE (SUBJECT TO CHANGE)				
Date		**Topic(s)**	**Key Deliverables**	
Thursday	**Monday**		**Thursday**	**Monday**
Week 1		Course Introduction Software Installation		
Aug 20	Aug 24			
Week 2		LLM Fundamentals: Models, APIs, and Interfaces		
Aug 27	Aug 31			
Week 3		Prompt Engineering AI Coding Agents and Vibe Coding		
Sep 3	Sep 14			
Week 4		Serving an Application Debugging and GitHub	**Assignment 1**	
Sep 10	Sep 21		Sep 17	Sep 28
Week 5		Context Engineering Retrieval-Augmented Generation (RAG)		
Sep 17	Sep 28			
Week 6		**Quiz 1** and Assignment 1 Review Hands-On Lab	**Assignment 2**	
Sep 24	Oct 5		Oct 5	Oct 16
Week 7		Multimodal Generative AI	**Project Proposal**	
Oct 1	Oct 12		Oct 11	Oct 22
Week 8		**Quiz 2** and Assignment 2 Review Hands-On Lab	**Assignment 3**	
Oct 15	Oct 19		Oct 22	Nov 2
Week 9		Fine-Tuning LLMs: SFT and PEFT (LoRA, QLoRA) Testing and Evaluation		
Oct 22	Oct 26			
Week 10		**Quiz 3** and Assignment 3 Review Hands-On Lab	**Assignment 4**	
Oct 29	Nov 2		Nov 5	Nov 9
Week 11		Security, Privacy, and Governance		
Nov 5	Nov 9			
Week 12		Assignment 4 Review Latest Developments in Generative AI		
Nov 12	Nov 16			
Week 13		Project Teamwork		
Nov 19	Nov 30*			
FALL BREAK				
Week 14		Final Project Presentations (Online Students Submit via Canvas)		
Dec 3	Dec 4†			

Final Exam Due: TBA

Notes

- *This Nov 30 class is on the Monday after fall break.
- †The final Friday of the semester (Dec 4) follows a Monday schedule and will meet in WLAW 305.
- There is no Monday class on Sep 7 (Labor Day) and no Thursday class on Oct 8 (Reading Day).
- Online students will follow the schedule in the second column (Week 1 beginning Aug 24).